

1a)

$$d = 0.30 \text{ in}$$

$$A = \frac{\pi d^2}{4} \rightarrow A = \frac{\pi (0.30)^2}{4} \rightarrow \boxed{A = 0.07069 \text{ in}^2}$$

1b)

$$F = 400 \text{ lbf}, E = 10 \times 10^6 \text{ psi}, \delta = 0.009 \text{ in}$$

$$\delta = \frac{FL}{AE} \rightarrow L = \frac{\delta AE}{F}$$

$$L = \frac{(0.009)(0.07069)(10 \times 10^6)}{(400)} \rightarrow \boxed{L = 15.90 \text{ in}}$$